



TMNA IT Experiences Process

DATA ENABLEMENT AND GLOBAL COMPLIANCE METADATA PROCESS

Purpose The purpose of the Metadata process is to document set of guidelines to meet and exceed the information needs of all the stakeholders in terms of information availability, consistency, and quality.

The goal of the metadata management is providing a consolidated, 360-degree view of information about important business parties, roles and products, particularly for more effective reporting and better impact analytics.

The Data Catalog enables Business and IT users realize the full potential of their enterprise data assets by improving business analytics, business context, user annotations, relationships and data governance by providing a unified view of enterprise information assets

Data Enablement and Global Compliance is the group that manages Data Governance at Toyota Motor North America (TMNA). Data Governance is the function to improvise, control, plan, monitor and enforce data management policies. Data Stewards will guide all the data management functions that are performed under the Data Enablement and Global Compliance program.

The goal is to deliver a Data Enablement and Global Compliance framework for TMNA to enable trusted, consistent and well defined information which drives smarter business decisions across the entire information supply chains.

Metadata is one of the key areas managed by Data Enablement and Global Compliance to ensure superior and efficient data assets. Data Enablement and Global Compliance will be an on-going program to kaizen data assets.

Scope The scope of this Metadata Process document is at the level defined below:

- Define the roles and responsibilities for different data usage and establish clear lines of accountability
- Develop best practices for effective data management
- To align data management efforts and technology with business needs
- To manage information consistently across the enterprise.

Overall Goals and Objectives of Metadata

- **Business Glossary** - Central repository for the business-related terms and

build governance to ensure a single definition of a business term across the enterprise.

- **Business Glossary Accessibility** - Business Glossary available to a larger set of audience within the organization through Intranet or Desktop access
- **Data Lineage** - End to end flow of data right from the source of generation to the target systems through the entire data integration environment.
- **Data Catalog** -Technical Metadata will be made available as a Catalog of Metadata which can be navigated, filtered and searched.

Compliance Compliance with this process is mandatory. When not meeting this process's requirements for business reasons, a TMNA manager must seek and obtain an approved Exception, requested and documented through the Exception process in [Toyota's Ticketing Systems](#). Team Members who violate this process may be subject to corrective action, up to and including termination. You are subject to all agreements you signed with the Company, as well as all Company rules and policies, including but not limited to the [TMNA Policies](#).

If you see or hear of an activity that may violate this process, whether by a Team Member, Corporate Partner, or Third-Party; you must report that activity. If this is a security incident you can email Security@toyota.com or call the Help Desk at (888) 488-4357. Reports may be made anonymously via the Toyota Speak Up Line by calling (844)SPEAK-99 or by visiting the [Toyota EthicsPoint website](#).

Owner	DataEnablement@toyota.com
Approver	DataEnablement@toyota.com
Initial Approval Date	08/30/2017
Effective Date	08/30/2017
Revision Date	10/15/2021
Last Reviewed Date	10/15/2021
Revision Number	1.0
SharePoint Location	DE Microsite (toyota.com)

NEVER USE PRINTED COPIES OF ANY PROCESS; USE SHAREPOINT VERSION.

This process contains the following categories:

1.	Processes	3
2.	Roles and Responsibilities.....	3
3.	References, Regulations, and Guiding Standards	5
3.1.	Metadata Lifecycle	5
3.2.	Data Domains	6

3.3.	Active Directory Groups	6
4.	Revision History	7
5.	Appendix for Relevant Processes	7

1. Processes

Applications not part of Data Enablement and Global Compliance Program.

Item	Requirement	Ref/Flag
1	Metadata process may not apply for data that is not part of Data Enablement and Global Compliance program	

Applications part of Data Enablement and Global Compliance Program.

Item	Requirement	Ref/Flag
1	All the data on-boarded on Toyota Big Data Platform (TBDP), its data sources, traditional and legacy platforms will be part of Metadata Management	
2	Metadata should be integrated with Business Glossary (Catalog), Technical Glossary and Data lineage platform for all data assets across North America.	
3	Business glossary, Technical Glossary and Data Lineage defined at tables, views and attributes level in Metadata should be reviewed and approved by data steward or business owner.	
4	As a part of data enrichment all the data assets of metadata should be classified for their confidentiality and should be labeled with appropriate data domains.	
5	Access to the metadata, technical and business glossary should be controlled through Active Directory (AD) groups and all the active directory groups created for metadata access should be reviewed and approved by Data Stewards.	
6	All the data access requests for metadata, technical and business glossary should have a One Toyota service (1TS) request number associated with it. No access should be granted to any user by e-mail, messenger or phone call request.	
7	Changes to the Process shall be reviewed and approved by Data Stewards and Metadata Owner.	
8	Metadata Process Changes will be defined by TMNA processes for Team Members and TMNA's Suppliers.	

2. Roles and Responsibilities

Role	Responsibility
Data Steward	▪ Represent departmental and enterprise data interests ▪ Sponsor and oversee data management projects and services ▪ Review and approve estimates of data asset value. ▪ Communicate and promote the value of information. ▪ Review and approve Business definitions/glossary for entities and attributes ▪ Monitor and enforce data policies and practices within a department.
Business Owner/Subject Matter Expert (SME)	▪ A knowledge worker and business leader recognized as a subject matter expert who is assigned accountability for the data specifications. ▪ Identify and define local and enterprise information needs. ▪ Propose, draft, review, and refine business names, definitions, and other data model specifications for assigned entities and data attributes. ▪ Responsible for evaluating and determining the appropriate confidentiality level for data.
Metadata Specialist	▪ Responsible for capturing and modeling data requirements, data definitions and business rules. ▪ Responsible for integration, control, and delivery of meta-data, including administration of meta-data repositories. ▪ Analyze and organize business entities, relationships, attributes, definitions and reference values. ▪ Establish data lineage and technical metadata for all applications. ▪ Enrich and categorize Business and technical glossary across all domains. ▪ Centralized Metadata management interface for Business and Technical Data ▪ Automated collection of all Metadata and linked Technical and Business Metadata views

Role	Responsibility
Metadata Developer	- Design and develop Metadata business rules - Perform necessary unit testing and participate in deploying scripts, dev and PROD environment as needed - Resolve the support issues or perform the support tasks on different environments - Monitor issues and highlight findings

3. References, Regulations, and Guiding Standards

3.1. Metadata Lifecycle

Item	Metadata Lifecycle	Description
1	Define	Identify and Define terms, classify it as per the confidentiality and label data assets with right domains.
2	Review	Review business definitions, functional definitions, classifications and domain for each data assets and get approval from Data Stewards.
3	Data Catalog	Technical Metadata needs to be made available as a Catalog of Metadata that can be navigated, filtered and searched
4	Data Lineage	Capture end to end flow of data right from the source of generation to the target systems through the entire data integration environment
5	Business Glossary	Enables Semantic Search, maintain a central repository for the business-related terms and build governance to ensure a single definition of a business term across the enterprise.

Item	Metadata Lifecycle	Description
6	Impact Analysis	Change management of upstream and downstream impact to a data asset.

3.2. Data Domains

All the data assets in Metadata should be classified as per the data domain listed below.

Item	Domain
1	Corporate Identity
2	Customer
3	Dealer Identity
4	Dealership
5	Digital Assets
6	Enterprise Corporate Applications
7	Finance
8	Identity Access Management
9	Manufacturing
10	Manufacturing Parts
11	Product Quality
12	Service History
13	Service Parts
14	Telematics
15	Vehicle
16	Warranty

3.3. Active Directory Groups

The below table provides high level active directory groups that will be created in metadata tool. The

number of AD groups may vary as per an application.

Item	Role	Permission		
		DEV	QA	PROD
1	Consumer/User	Read	Read	Read
2	Data Admins	Read Write Execute	Read Write Execute	Read Write Execute
3	Data Analyst	Read Execute	Read Execute	Read
4	Data Architects	Read	Read	Read
5	Data Owners/Business Owners	Read	Read	Read
6	Data Stewards	Read Execute	Read Execute	Read Execute
7	Developer	Read Write Execute	Read	Read

4. Revision History

Version	Date	Author	Description	Section(s) Affected
1.0	08/30/2017	DataEnablement@toyota.com	The Metadata Process document define metadata rules and policies for ownership, standardization and security	All
1.0	10/15/2021	DataEnablement@toyota.com	The Process was moved to new template	All

5. Appendix for Relevant Processes

Data Enablement ITx Data Quality Process

[TMNA_DE_Data_Quality_Process.pdf \(toyota.com\)](#)